# Year 12 Pre-Mock Recap — cumulative 1.1–1.4 + 2.1 + 2.2.1

Name: ______________________ Date: __________ Mark: ______ / 60

**Time allowed: 75 minutes. Closed book.**

*OCR H446 cumulative pre-mock recap. Section A covers Component 1 (36 marks); Section B covers Component 2 (24 marks). Show all working.*

## Questions

### Section A — Component 1 (36 marks)

**Q1.** Describe the role of the **address bus**, the **data bus** and the **control bus** when the CPU fetches an instruction from main memory. **[3]** — *1.1.1*

**Q2.** A desktop computer contains both **RAM** and **ROM**. (a) State **two** differences between RAM and ROM. **[2]**

(b) Give **one** reason the computer needs ROM. **[1]**

**[3 total]** — *1.1.3*

**Q3.** An operating system is managing three processes. (a) Describe how **round robin** scheduling shares the processor between the processes. **[2]**

(b) Explain how a long job could be treated unfairly under **shortest job first** scheduling. **[1]**

(c) State the name of the memory-management technique that divides memory into **fixed-size** blocks. **[1]**

**[4 total]** — *1.2.1*

**Q4.** A developer writing firmware for a microcontroller programs in **assembly language**. (a) State the name of the translator that converts assembly language into machine code. **[1]**

(b) Give **one** reason assembly language is appropriate for this task. **[1]**

(c) Within a machine-code instruction, state what is meant by the **opcode** and the **operand**. **[1]**

**[3 total]** — *1.2.4*

**Q5.** A student types `www.revisionhub.co.uk` into a web browser. (a) Describe how **DNS** is used to obtain the IP address of the web server, including what happens when the local resolver does **not** already know the address. **[3]**

(b) State what is meant by a **protocol**, and give **one** reason standard protocols are essential on the internet. **[2]**

**[5 total]** — *1.3.3*

**Q6.** Show all working. (a) Convert the denary number **154** into 8-bit binary. **[1]**

(b) Convert **154** into hexadecimal. **[1]**

(c) Using **8-bit two's complement**, calculate **47 − 61**: show 47 and 61 in 8-bit binary, form the two's complement of 61, then perform the addition and state the final **denary** result. **[3]**

**[5 total]** — *1.4.1*

**Q7.** Show each step. (a) Use **De Morgan's law** to rewrite **¬(B ∨ ¬C)** so that NOT is no longer applied to a bracket. **[2]**

(b) Simplify **A∧B ∨ A∧B∧C**, naming the law used. **[2]**

**[4 total]** — *1.4.3*

**Q8.** A sorted collection of values is stored in a **linked list**: `14 → 26 → 33`. (a) Describe how a linked list stores its items, referring to **nodes** and **pointers**. **[2]**

(b) Describe the steps needed to insert the value **29** into its correct position, and state why no existing data has to be moved. **[2]**

**[4 total]** — *1.4.2*

**Q9.** A music festival issues electronic wristbands. Battery-powered scanners at each gate currently check every wristband against a **central database** over the site's Wi-Fi; the organisers are considering instead giving each scanner its **own local copy** of the ticket data. Evaluate the two approaches, referring to **data consistency**, **reliability of the network link**, and the **security of transmitted data**. **[5]** — *synoptic 1.3.1 / 1.3.2 / 1.3.3*

## Section B — Component 2 (24 marks)

**Q10.** A city introduces a bike-hire scheme: riders unlock a bike at a docking station using an app, ride it, and return it to any station. The app shows a map of stations with the number of bikes available at each.
(a) Identify **one** example of **abstraction** in the app's map. **[1]**

(b) The developers **decompose** the overall problem. Identify **two** sub-problems of "hire a bike". **[2]**

(c) Identify **one** part of the system that is well suited to **concurrent** processing. **[1]**

**[4 total]** — *2.1.1 / 2.1.3 / 2.1.5*

**Q11.** A toll-bridge barrier rises only when payment has been accepted **and** the vehicle's height, held in the variable `height`, does not exceed **2.4** metres. The Boolean variable `paymentOK` is available. (a) *Thinking logically:* write the **Boolean condition** the program should test before raising the barrier. **[2]**

(b) *Thinking ahead:* the toll charge looked up for each regular number plate could be **cached** after its first lookup. State **one benefit** and **one risk** of this caching. **[2]**

**[4 total]** — *2.1.4 / 2.1.2*

**Q12.** The following program is written in OCR Exam Reference Language.

```
n = 305
rev = 0
while n > 0
    rev = rev * 10 + (n MOD 10)
    n = n DIV 10
endwhile
print(rev)
```

(a) Complete the trace table, showing the values of `n` and `rev` at the **end of each iteration** of the loop. **[3]**

| Iteration | n | rev |
|---|---|---|
| start | 305 | 0 |
| 1 | | |
| 2 | | |
| 3 | | |

(b) State the value output by the program. **[1]**

(c) State the purpose of the program. **[1]**

**[5 total]** — *2.2.1*

**Q13.** A swimming coach records a swimmer's lap times, in seconds (real numbers), in the array `laps`. Write a function `countFaster(laps, target)` in **OCR Exam Reference Language** that: - returns **−1** if `laps` is empty; - otherwise returns the **number of laps** swum in **strictly less than** `target` seconds. Use `laps.length` to find the number of items in the array. **[6]** — *2.2.1*

**Q14.** A programmer is improving a menu-driven program. (a) State **two** features of an **IDE** that help the programmer find the cause of a logic error, describing how each helps. **[2]**

(b) Give **one** reason for using a **named constant** rather than repeating a literal value through the code. **[1]**

(c) The menu must be displayed **at least once**, and redisplayed until the user enters a valid choice. Name the most suitable **iteration structure** in OCR Exam Reference Language and justify your choice over `while ... endwhile`. **[2]**

**[5 total]** — *2.2.1*